Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 12 GB of memory. The learning rate was set to 0.001 with the Adam optimizer.

2 Results

Figure 1 shows the validation accuracy during training.

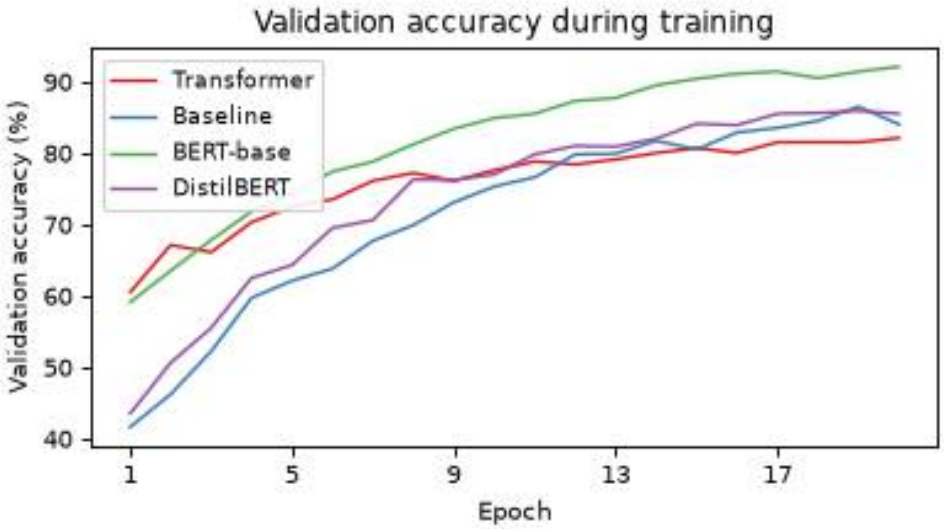


Figure 1: Validation accuracy during training for the models in this experiment.

DistilBERT attains a validation accuracy of 90.4% at epoch 15. Baseline attains a validation accuracy of 86.5% at epoch 19. As shown in Figure 1, Transformer reaches 66.5% validation accuracy at epoch 10. Baseline attains a validation accuracy of 46.2% at epoch 2.

Figure 2 shows the test accuracy during training.

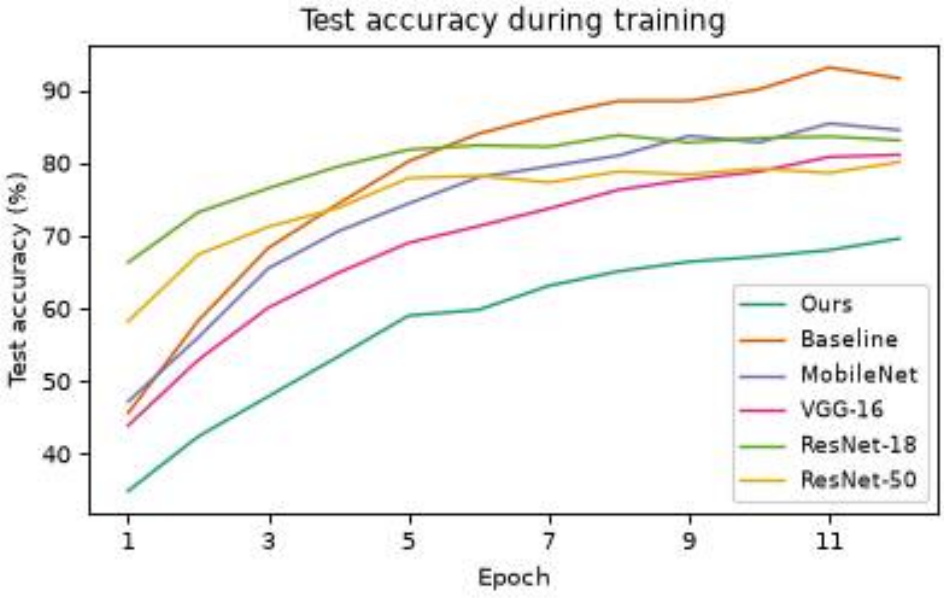


Figure 2: Test accuracy during training for the models in this experiment.

VGG-16 attains a test accuracy of 79% at epoch 10. At epoch 2, ResNet-50 is 15 points better than VGG-16 (Figure 2). VGG-16 outperforms Ours by 5.8 percentage points at epoch 3. From epoch 3 to epoch 5, the test accuracy of VGG-16 improves by a relative 14.8% (Figure 2).

Figure 3 shows the training loss.

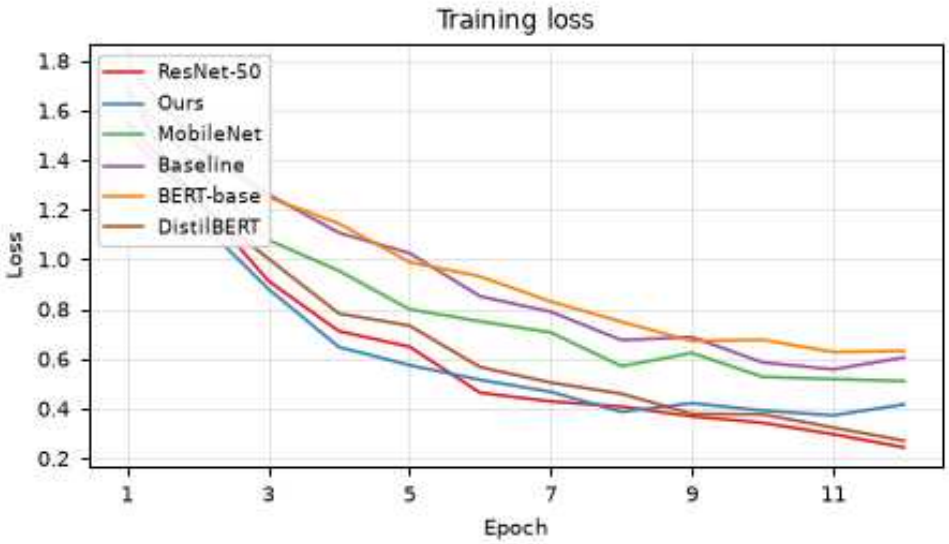


Figure 3: Training loss for the models in this experiment.

As shown in Figure 3, DistilBERT has a loss of about 0.4 at epoch 10. As shown in Figure 3, at epoch 12, the loss of MobileNet is 0.425. At epoch 10, the loss of Ours is 0.342 (Figure 3).

3 Discussion

On STL-10, Logistic Regression obtains an accuracy of 82.8%. Overall, the results are in line with what we expected from the literature.